

Loan Approval Prediction — Exploratory Data Analysis (EDA)

Financial Data Analysis: Credit History, Income, Loan Amount, Education & Employment vs Approval

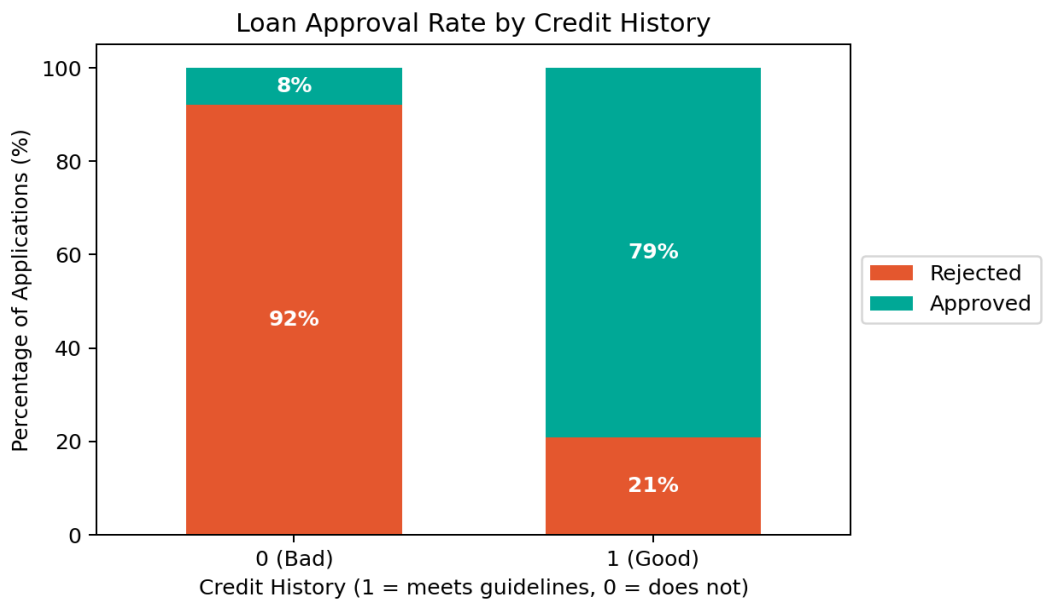
1. Approval Analysis

Factor	Group	Approval Rate
Credit History	Good (1)	79.0%
Credit History	Bad (0)	7.9%
Education	Graduate	71.0%
Education	Not Graduate	60.7%
Self Employed	No	68.8%
Self Employed	Yes	68.3%

Median Total Income — Approved: 5,439 vs Rejected: 5,289.5 (nearly identical). Median Loan Amount — Approved: 128 vs Rejected: 128 (identical). This shows income and loan amount alone barely separate approved from rejected applicants.

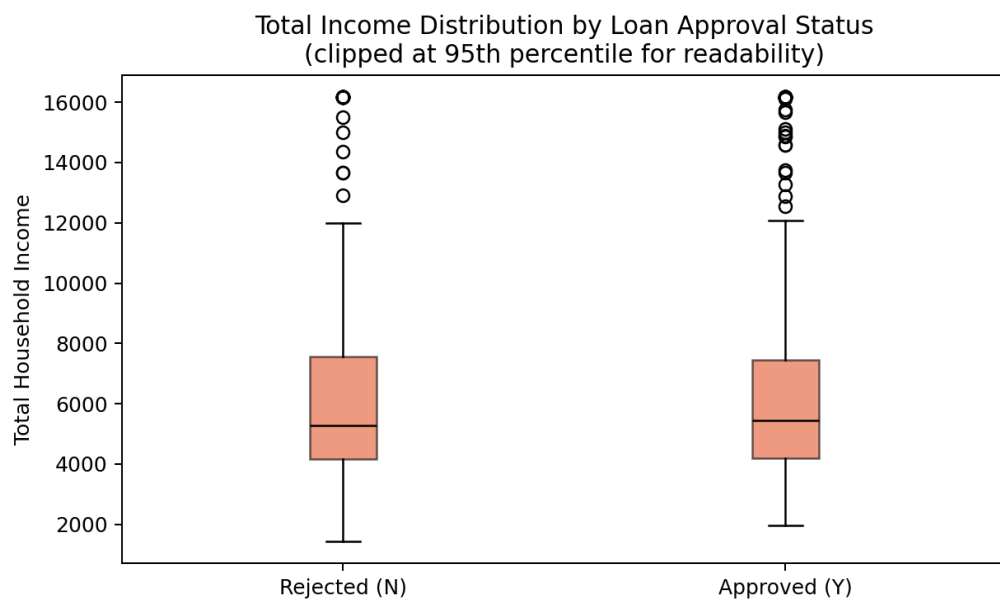
2. Visualizations

2.1 Approval Rate by Credit History



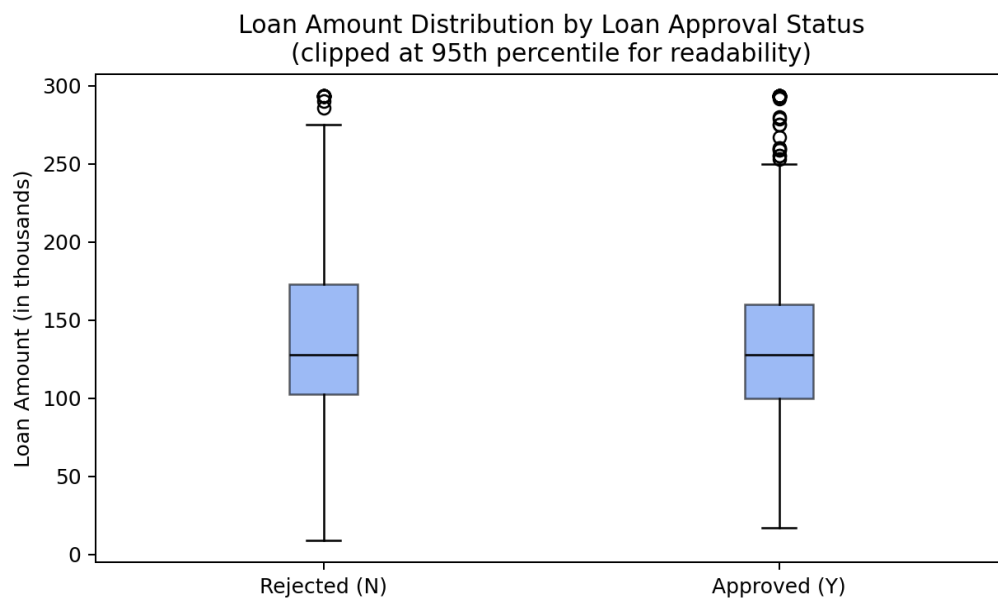
Applicants with a good credit history (1) are approved 79% of the time, versus only 8% for those with a bad credit history (0) — the single strongest factor in the dataset.

2.2 Total Income vs Approval Status



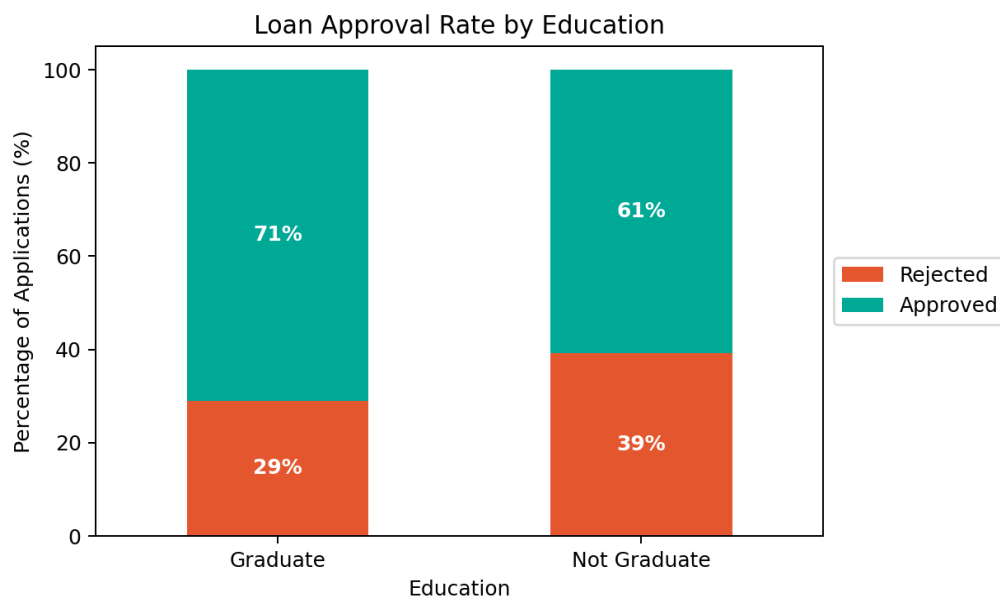
Income distributions for approved and rejected applicants overlap heavily, with very similar medians. Income alone is a weak predictor of approval in this dataset.

2.3 Loan Amount vs Approval Status



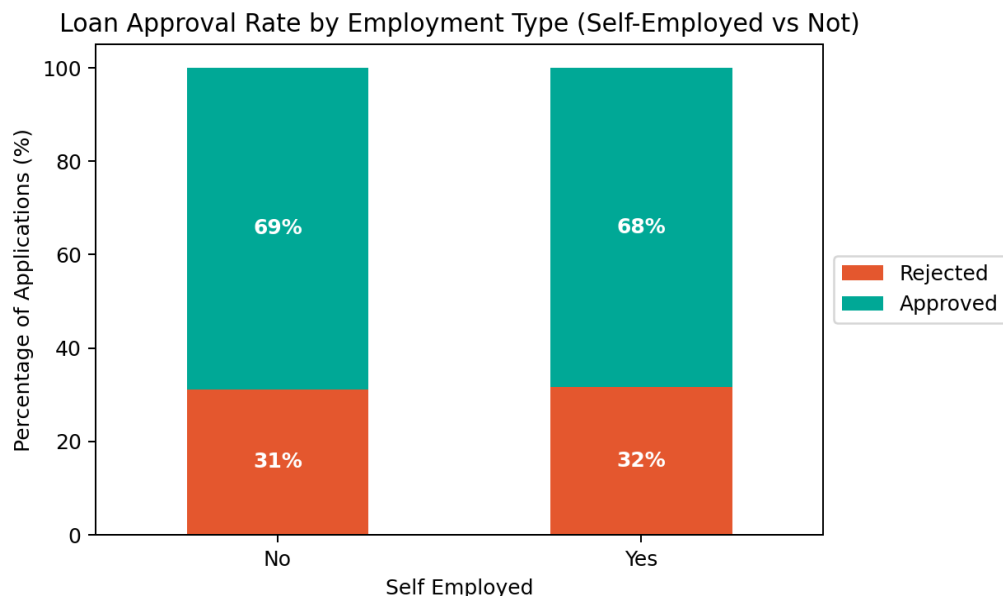
Loan amount distributions are also nearly identical between approved and rejected groups, suggesting the requested amount by itself does not strongly drive the approval decision.

2.4 Approval Rate by Education



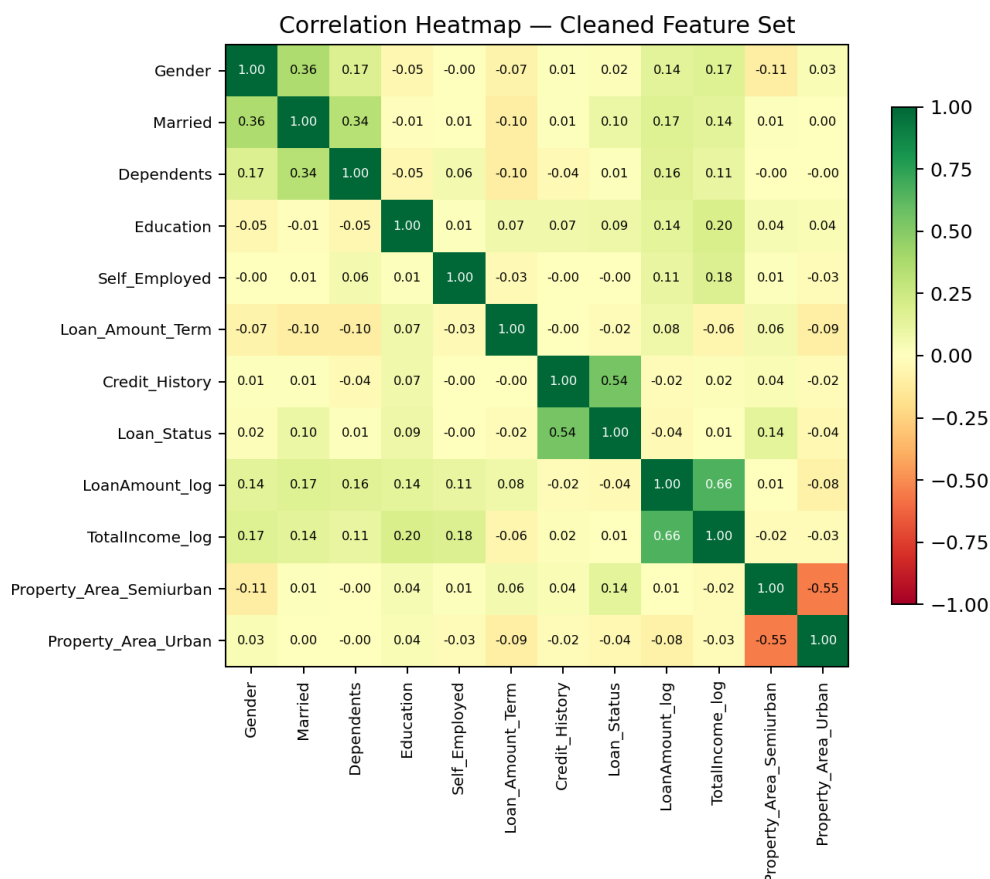
Graduates are approved slightly more often (71.0%) than non-graduates (60.7%) — a mild positive relationship.

2.5 Approval Rate by Employment Type



Self-employed and salaried applicants are approved at almost the same rate (68.3% vs 68.8%) — employment type shows virtually no relationship with approval.

2.6 Correlation Heatmap (All Cleaned Features)



Credit_History ($r = 0.54$) is by far the strongest correlate of Loan_Status. All other features correlate weakly ($|r| < 0.15$) with approval, though LoanAmount_log and TotalIncome_log correlate strongly with each other ($r = 0.66$), as expected since larger loans tend to accompany higher incomes.

3. Correlation / Relationship Analysis

Feature	Correlation with Loan_Status
Credit_History	+0.54 (strong positive)
Property_Area_Semiurban	+0.14 (weak positive)
Married	+0.10 (weak positive)
Education	+0.09 (weak positive)
Gender	+0.02 (negligible)
Dependents	+0.01 (negligible)
TotalIncome_log	+0.01 (negligible)
Self_Employed	-0.00 (negligible)
Loan_Amount_Term	-0.02 (negligible)
LoanAmount_log	-0.04 (negligible)
Property_Area_Urban	-0.04 (negligible)

Interpretation: Only Credit_History shows a meaningful linear relationship with loan approval. Financial factors like income and loan amount, taken alone, are not strongly correlated with the outcome — this suggests approval decisions depend more on repayment reliability (credit history) than on the raw size of income or loan requested.

4. Important Observations

- Credit History is the dominant driver of loan approval — applicants with a good credit history are approved ~10x more often (79% vs 8%) than those without one.
- Income and Loan Amount, on their own, show almost no difference between approved and rejected applicants (near-identical medians) — meaning the bank does not simply approve 'high income, low loan' applicants.
- Education has a mild positive relationship with approval (Graduates: 71% vs Not Graduate: 61%).
- Employment type (Self-Employed vs Salaried) shows virtually no effect on approval outcome.
- LoanAmount and TotalIncome are strongly correlated with each other ($r = 0.66$) — larger loans are requested by higher-income applicants, as expected, but this pair still doesn't strongly predict approval.
- Since Credit_History has the most predictive power, it will likely dominate the Logistic Regression model's coefficients — consistent with what was already seen when the model was trained.